BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 Data Analysis Report

Project ID: **25-26J-088**

Project Title: **AI-BASED EARLY CHILDHOOD EMOTIONAL INTELLIGENCE MONITORING PLATFORM IN URBAN AREA**

1. Introduction
   1.1. Background:

For early childhood emotional intelligence is crucial because lower emotional intelligence impacts clear thinking process and the education of the preschoolers [1].Lower emotions mean emotional dysfunctional. By identifying the children that are emotionally distressed earlier on, we can help them before the issue gets worse. Children maybe going through varies emotional challenges at this stage and might be hiding their true emotions. Children that are heavily impacted to these emotional changes are children who have divorced parents, children who has lost parents, children coming from violent backgrounds. From ages 4-6 children start to recognize emotions and label them correctly [2-4]. When a small child is not give the space to express their feeling, they start to lock them self and adopts an attitude of withdrawal [5]. After wards these emotions starts to become uncontrollable reactions. The three main emotions that they can be properly identified at the age of 4-6 are anger, sad and happy [2-4]. Neglecting a child's emotional needs, negatively impacts early childhood development, especially for children who cannot express their feelings clearly[6]. The main problem the research is trying to solve is to identifying the emotions in early childhood children. The focus of this study will be children aged 4-6 in Urban areas, Sri Lanka. Prior applications that were build for the early childhood, gives the child a digital device [7-10], which increase digital addiction in children.

Emotions are connected to the brain through the amygdala and emotions affects the child's cognitive functions such as memory recall, attention and spoken language in different ways. Hippocampus is the main brain part that is connected to memory recall. Positive emotions strengthen long-term memory and information processing, while negative emotions such as fear, anxiety, and chronic stress impair memory formation, reduce attention, and disrupt recall accuracy [11]. The anterior cingulate cortex (ACC), serves as a critical node connecting attention and emotional intelligence.Negative emotions can capture and hold attention, making it difficult for children to shift focus or sustain attention on tasks deemed non-emotional or less salient. This directly impacts learning and engagement in structured activities. Spoken language skill is connected to the brain with broca's area [12]. When children experience anxiety or poor emotional regulation, they exhibit grammatical errors, tense inconsistencies, and topic drift due to reduced prefrontal control over Broca's area, which governs grammatical structuring and speech planning. Researches shows that emotion regulation directly affects pragmatic skills such as topic relevance, turn-taking, and coherence, because emotional dysregulation consumes cognitive resources needed for structured language production.

.

Early childhood education frameworks such as UNESCO Early Childhood Care and Education (ECCE), WHO–UNICEF *Nurturing Care Framework* and OECD Early Learning Frameworks state that preschools should observe children's emotional functioning, identify children who may need additional support, engage caregivers and specialists when concerns arise. This application helps preschool and the parents monitor these through the child's daily activities.

1.2. Research Problem:

Emotional dysfunctionality, which reflects low emotional intelligence, can significantly disrupt a child's cognitive, social, and psychological growth if it is not identified and addressed at an early stage. However, detecting emotional dysfunctionality in young children remains a major challenge, particularly when children lack the verbal ability or emotional awareness to clearly express their internal emotional states.

In many existing approaches, emotional assessment relies heavily on digital technologies such as mobile applications, facial recognition systems, or wearable devices. While these methods may offer technical advantages, they raise serious concerns regarding child privacy, ethical data usage, and cultural acceptability. Furthermore, constant digital exposure may itself negatively influence a child's emotional development. Therefore, there is a clear need for a non-digital, privacy-preserving, and culturally appropriate method to identify emotional dysfunctionality in young children, especially within the Sri Lankan context.

Analyzing emotional dysfunctionality during early childhood is crucial because this developmental stage forms the foundation for lifelong emotional regulation and interpersonal skills [13]. When young children are unable to verbally articulate their feelings, they often internalize their emotions, leading to social withdrawal, reduced communication, and emotional isolation [5]. Over time, these suppressed emotions may evolve into uncontrollable emotional reactions, behavioral problems, anxiety, or aggression. If such emotional patterns are not identified early, they may persist into adolescence and adulthood, resulting in long-term psychological and social consequences.Neglecting a child's emotional needs can significantly hinder early childhood development, particularly for children who struggle to communicate their emotions clearly [14].

The impact of emotional dysfunctionality is even more severe among vulnerable child populations. Foster children, children who have recently lost a parent, refugee children, children with special needs, adopted children, and children from violent or unstable family environments are at a much higher risk of emotional distress. These children often experience trauma, insecurity, and inconsistent emotional support, which directly influence their emotional regulation abilities. Without appropriate early intervention, such children may develop long-term trust issues, social withdrawal, academic difficulties, and behavioral disorders.

Therefore, the core research problem is the absence of a reliable, privacy-preserving, non-digital, and culturally relevant method to identify emotional dysfunctionality in young Sri Lankan children. Addressing this problem is essential to support early intervention, improve emotional well-being, enhance educational outcomes, and promote healthier social development. By providing an effective early identification mechanism, this research aims to contribute to safeguarding children's emotional health while respecting their dignity, privacy, and cultural identity.

1.3. Objectives:

It Number	Objective	Objective number
IT22315250	Identification of emotions of children aged 4-6, while respecting their privacy in an indirect, reliable mechanism	1
IT22345400	Identification of language skills verbally using an AI based voice assistant	2
IT22640048	Identification of fine motor level & attention level of children aged 4 to 6	3
IT22358066	Identification of memory recall strength using quantitative scoring methods	4

2. Data Exploration

2.1. Data Collection

Below mentioned are the two methods that we used to collect our datasets.

- **Educational institution**s

Data collection was conducted through visits to multiple preschools after obtaining an official approval letter from SLIIT. This letter enabled us to approach educational institutions and request permission to collect the required data for all components of the study. Ethical and administrative approvals were granted by the respective institutions prior to data collection.

The educational institutes are:

- HummingBird Learning center and day care malabe and kahanthota branches
- Christine Alexandria Activity School and Daycare Centre
- Bomiriya Kanishta school
- Leeds International School
- Brightwood International School

- **Publicly available datasets from kaggle**

Finding emotions through art work:
https://www.kaggle.com/datasets/serdarciftci/kido-children-drawing-dataset?resource=download

Custom created dataset was used with the Kaggle Grammar Error Correction dataset:
https://www.kaggle.com/datasets/satishgunjal/grammar-correction

Custom dataset created using publicly available online sources to represent animals, fruits, and vegetables aligning with the designed recall tasks.

2.2. Dataset Description:

Data source	Description	Resource	Size	Key attributes
Preschool artwork	Dataset was mainly collected by visiting the preschools	Preschool institutions, kaggle: https://www.kaggle.com/datasets/serdarciftci/kido-children-drawing-dataset?resource=download	600 data was labeled into emotions	Class labels: The data was labeled into three classes, anger, happy, sad with the help of an art therapist.
Emotional regulation checklist	Dataset was mainly collected by visiting the preschools	Preschool institutions	300 rows of data	Classes labels: The dysfunctional levels can be calculated using this checklist. The levels are low, medium and high. These were labeled with the help of this research paper [15].
Audio recordings	Data was mainly collected by telling the children to repeat the sentence in storybooks after me.	Preschools /schools	467 audio recordings (Totally 2 hours of data)	Audio recordings were collected in .wav 16kH format using a laptop in a low noise area. Labels: Transcriptions were manually created and double checked to consider them as the ground truth.

Preschool Grammar error detection and correction	Dataset was created to identify common grammar errors done in spoken language.	Generated data, Kaggle: https://www.kaggle.com/datasets/satishgunjal/grammar-correction	6300 rows of data	Column names: Serial number, Error Type, Ungrammatical sentence, Standard English. Error type column has 9 error types which are identified as common error types done by preschoolers. Kaggle dataset was used by removing unnecessary error types. Using that dataset new data was generated and validated by a preschool English teacher.
Preschool coloring activity	Dataset was mainly collected by visiting the preschools	Preschool institutions	179 activity images	Computer vision and image preprocessing techniques were applied to the coloring activity images to extract fine motor–related features and attention-level features, which were used to generate scores for both components.
Preschool tracing activity	Dataset was mainly collected by visiting the preschools	Preschool institutions	62 activity images	Computer vision and image preprocessing techniques were applied to the coloring activity images to extract fine motor–related features and attention-level features, which were used to generate scores for both components.
Preschool handwriting activity	Data were collected by conducting handwriting-based memory recall activities where children reproduced previously learnt letters	Preschool institutions	7712 handwriting samples	Character Labels: The handwriting data consists of Sinhala letters (ඉ, ශ, ඥ, ඤ) and English letters (m, f, j, M, F, J)
Preschool categorization activity	Data were collected for categories that are familiar to preschool children to support categorization-based	Publicly available online sources (Google Images):	1740 relevant object images	The dataset is organized into three main categories; Animals, Fruits, and Vegetables and includes labeled samples covering 32 specific animal species (such as dog, cat, and bear), 12 vegetable types, and 13 fruit types.

	memory recall activities			

2.3. Suitability Analysis

2.3.1. Relevance to Individual Research Objectives:

The datasets perfectly align with each of the components objectives. All the data collected are preschool datasets

	1	2	3	4
Data source 1	x			
Data source 2	x			
Data source 3		x		
Data source 4		x		
Data source 5			x	
Data source 6			x	
Data source 7				x
Data source 8				x

3. Methodology

3.1. Data Preprocessing:

Objective 1: IT22315250

	Transformation technique					
	Data cleaning	Data normalization	Data augmentation	Feature engineering	Data Encoding	Dimensionality reduction (PCA)
Data source 1	X	X	X	X		X
Data source 2	X		X		X	

Objective 2: IT22345400

	Transformation technique								
	Data Cleaning	Handling Missing Data	Data Type Conversion	Data Normalization	Feature Engineering	Data Encoding	Data Padding & Masking	Data Truncation	Data Shufflin g
Data source 3	x	x	x	x	x	x	x	x	x
Data source 4	x	x			x				

Objective 3: IT22640048

	Transformation technique					
	Data Cleaning	Data Normalization	Feature Engineering	Data Encoding	Edge Detection	Morphological Processing
Data source 5	x	x	x	x	x	x
Data source 6	x	x	x	x	x	x

Objective 4: IT22358066

	Transformation technique						
	Data cleaning	Data normalization	Noise removal	Data augmentation	Data Scaling	Data Discretization	Feature engineering
Data source 7	X	X	X	X	X	X	X
Data source 8	X			X	X		

3.2. Scalability

Data source 1:
Due to the limited size of the primary dataset, data augmentation techniques are applied to enhance data diversity and improve model generalization. Deep learning models are selected as they are capable of learning complex feature representations even when trained on relatively small datasets, particularly when supported by augmentation strategies. However, if the model demonstrates difficulty in achieving satisfactory learning performance, an additional publicly available dataset from Kaggle will be incorporated. Prior to its use, all labels in the Kaggle dataset will be reviewed and validated by a qualified art therapist to ensure consistency and reliability with the research objectives. The selected Kaggle dataset has been widely used in related studies and is considered a reliable source for children's drawing analysis research. Kaggle:
https://www.kaggle.com/datasets/serdarciftci/kido-children-drawing-dataset?resource=download

Data source 2:
The dataset includes simplified rating scores that represent the child's level of emotional dysfunctionality. If the available data are insufficient for the model to effectively learn meaningful patterns, additional data will be collected from preschool teachers to improve dataset robustness. Given that the scoring process is well-defined and supported by established research literature, synthetic data generation techniques may also be employed to augment the dataset [15]. These synthetic samples will be generated based on the statistical characteristics of the real data to preserve validity and ensure consistency with the original scoring methodology.

Data source 3:
The current audio dataset, consisting of 467 speech recordings paired with corresponding transcriptions, is sufficient for meaningful data analysis and exploratory assessment of child speech characteristics and transcription quality. This dataset size allows for reliable examination of common speech patterns, pronunciation variability, transcription accuracy, and basic distributional properties such as utterance length and vocabulary usage. The use of a consistent file-based structure (separate audio and transcription folders with one-to-one mapping) enables straightforward expansion, allowing additional recordings to be added without modifying the existing preprocessing or analysis code. So, while the current dataset is adequate for present analytical objectives, its structure and preparation pipeline are well-suited for deeper analysis as additional data becomes available.

Data source 4:
The dataset used for grammar error detection and correction consists of four key columns and is structured to support supervised learning for text-to-text transformation. Given the moderate number of error categories and the focused linguistic scope, the dataset size (6300 rows) is sufficient for fine-tuning a pre-trained transformer model such as T5, which benefits from transfer learning

even with limited task-specific data. Furthermore, the dataset is scalable, as new grammar error instances can be continuously generated and validated within the same schema, enabling incremental expansion to improve model robustness without altering the underlying data structure or training pipeline.

Data source 5:
The coloring dataset consists of 179 images, each representing an individual child's coloring performance. Although this dataset size is insufficient for training deep learning models due to the high risk of overfitting and poor generalization, even when data augmentation is applied, it is adequate for computer vision–based analysis. Each image contains rich spatial and structural information that supports objective analysis of fine motor and attention-related performance. Furthermore, the approach is scalable, as additional images can be incorporated without retraining, and computational complexity increases linearly with the dataset size.

Data source 6:
The tracing dataset consists of 62 images, each representing an individual child's tracing performance. This dataset size is insufficient for deep learning–based modeling due to its limited sample size and the resulting risk of overfitting, even with data augmentation. However, the structured nature of tracing tasks makes the dataset suitable for computer vision–based analysis. The method remains scalable, as new tracing samples can be processed independently without affecting system performance.

Data source 7:
With 7712 samples collected covering both Sinhala and English letters**,** the dataset size is sufficient for memory recall analysis, as it includes multiple handwriting variations per letter. To further enhance scalability, data augmentation techniques were applied to increase variation and robustness without requiring additional data collection. The relatively large number of handwriting samples supports scalable analysis by allowing consistent feature extraction and comparison across different letters and writing variations. Additionally, the dataset can be easily expanded in future by incorporating more handwriting samples without requiring changes to the preprocessing pipeline.

Data source 8:
The categorization dataset contains 1740 records across the categories of fruits, vegetables, and animals, which is adequate for scalable analysis of categorization-based memory recall. Data augmentation techniques were used to improve diversity and reduce bias, supporting better generalization as the dataset grows. The dataset design is compatible with pre-trained models**,** allowing scalable analysis without the need for very large datasets. Furthermore, the dataset can be easily extended by incorporating new categories or additional records while preserving the existing preprocessing and analysis workflow.

3.3. Feature extraction

Objective 1: IT22315250

Data source 1:

1. **Data cleaning and filtering:** After collecting the art works of the preschool children, they were shown to the art therapist, who helped identify the drawings that can be used for the research and which drawings that are not suitable for the research. All images were converted to RGB to ensure consistency. Images were resized and cropped to a fixed input size (224×224), ensuring uniformity across samples

2. **Feature extraction:** Data augmentation was applied before feature extraction to increase feature diversity. Feature extraction was performed using a pre-trained Vision Transformer model trained on ImageNet. Each image extracted a high dimensional feature vector that captures color, spatial relationships, texture.

3. **Feature Normalization:** This was done using the standardScale which will ensure the mean is 0. This is essential because after the feature extraction, over 2000 features were extracted and we need to perform a dimension reduction and models like SVM with RBF kernel relies on distance calculation.

4. **Feature transformation:** This was carried out with the help of the principal component analysis (PCA). This was able to reduce the original features to 150 components. It transformed correlated high dimensional features into uncorrelated principal components. It removed the redundant features while perceiving the distinct features. This will improve the models computational efficiency, noise reduction and model generalization.

5. **Feature selection:** PCA was done to capture the most relevant features while disregarding the noises from the features.LIME model was used for the art feature extraction and highlighting in the original image, helping the user trust the system more.

6. **Feature evaluation:** Feature qualities were evaluated indirectly through classification accuracy, cross validation accuracy, confusion matrix and ROC-AUC scores. High ROC and stable cross validation performance indicate that the extracted and transformed features effectively captured the emotional patterns.

Data source 2:

1. **Data labeling and cleaning:** The data was labeled using the emotional regulation checklist calculations that are shown in the research paper [15]. The target variable (dysfunction level) was converted into a structured ordinal format.

2. **Feature extraction:** Using predefined emotional regulation checklist questionnaire items as numerical feature inputs. Each feature corresponds to a psychometrically validated behavioral

indicator. No automated deep feature extraction was required since features were already structured and domain defined.

3. **Feature normalization:** This was done using the standardScaler.

4. **Feature selection:** Feature selection was conducted manually based on psychological validity of the emotional regulation checklist items and the prior literature confirming their relevance to emotional dysfunctionality. Domain driven feature selection was used [15].

5. **Feature evaluation:** Feature effectiveness was evaluated using training and testing accuracy, confusion matrix, precision, recall and f1 scores from the classification reports . These metrics confirmed whether the selected emotional regulation checklist features were predictive of dysregulation levels.

Objective 2: IT22345400

Data source 3:

1. **Dataset Loading and Verification:** The audio dataset, comprising 467 child speech recordings with corresponding transcriptions, was loaded from Google Drive with separate folders for audio and text files. Verification checks ensured that every audio file had a corresponding transcription, and any missing or empty transcriptions were identified and excluded. This step confirmed the completeness and integrity of the dataset. A few sample recordings were inspected to evaluate the representativeness of the speech content and transcription quality.

2. **Dataset Splitting and Distribution:** The dataset was shuffled and divided into training (373 samples), validation (46 samples), and test (48 samples) subsets. This split ensured that each subset contained a sufficient number of samples for training the model, monitoring performance, and evaluating generalization. The distribution of samples was carefully assessed to maintain consistency and representativeness across all subsets.

3. **Audio Standardization and Preprocessing:** All audio files were cast to a standardized format with a 16 kHz sampling rate, ensuring uniformity for feature extraction. The audio recordings were then converted to log-Mel spectrogram features using the Whisper feature extractor, while transcriptions were tokenized into integer token IDs and truncated to comply with model limits. Inspection of a few preprocessed samples confirmed that features and labels were correctly aligned and suitable for model training.

4. **Data Collation and Augmentation:** A custom data collator was used to dynamically pad input features and labels, ensuring batch compatibility during model training. Additionally, a framework for audio data augmentation (including time stretching, speed changes, and volume perturbations) was implemented to enhance dataset diversity and robustness,

although it was not applied in the current run. These steps supported the generation of high-quality feature representations for reliable model input.

5. **Evaluation of Dataset Features:** Finally, the preprocessed dataset enabled model evaluation using Word Error Rate (WER) on the validation and test sets. This assessment provided insight into transcription accuracy, highlighting the dataset's adequacy for ASR fine-tuning while identifying areas where additional data could improve performance. Overall, the dataset is sufficient for initial analysis and is structured to be scalable, allowing new recordings to be easily integrated for ongoing growth and deeper analytical studies.

Data source 4:

1. **Dataset Loading and Structure Inspection:** The dataset was first loaded from a CSV file, and its basic structure was examined, including the number of records, column names, and data types. This step ensured that all attributes Serial Number, Error Type, Ungrammatical Statement, and Standard English were correctly represented and suitable for a text-to-text grammar correction task. Inspecting the dataset shape and types also confirmed that the data was in a format compatible with transformer-based models.

2. **Categorical Feature Analysis (Error Type):** The Error Type attribute, which contains nine preschool-relevant grammar error categories, was evaluated for its distribution and balance. Descriptive statistics and visualizations such as count plots were used to identify the frequency of each error type. This analysis ensured that all categories had sufficient representation and that no error type would disproportionately influence the model, maintaining fairness and reliability in training.

3. **Data Quality Assessment:** Completeness and consistency of the textual attributes were checked. Missing values were identified, and it was confirmed that all Ungrammatical Statement and Standard English pairs were complete. Duplicate entries across key attributes were detected and removed to prevent bias and overfitting. Additionally, the Serial Number column was evaluated and found to be non-informative for the model, leading to its removal.

4. **Text Cleaning and Normalization:** Textual preprocessing was performed on the Error Type, Ungrammatical Statement, and Standard English columns. Steps included trimming leading and trailing whitespaces and normalizing internal spaces. This standardization ensured that inconsistencies in formatting would not interfere with tokenization or model learning.

Objective 3: IT22640048

Data source 5:

1.Image Preprocessing and Standardization: Load the input image from bytes or file-like object using numpy's frombuffer function, decode it with OpenCV's to convert the byte array into a usable image matrix, resize the image to a standardized size, and convert the color space from BGR to RGB for proper color representation in subsequent processing steps and then transform to HSV (Hue, Saturation, Value) color space which separates color information from intensity making it more robust for color detection under varying lighting conditions.

2.Color Detection and Template Boundary Extraction: Apply HSV thresholding with saturation range to create a binary mask isolating all colored regions made by the child, while simultaneously converting the image to grayscale and detecting template boundaries through combined dark line detection and Canny edge detection. Apply morphological closing operations for 2 iterations to connect broken boundaries, then use contour detection to identify the largest shape as the primary template with minimum area threshold, creating both a filled expected area mask and a boundary edges mask.

3.Completeness and Boundary Adherence Extraction: Perform bitwise AND between the colored mask and expected area mask to count filled pixels inside boundaries, divide by total expected pixels and multiply by 100 to extract the completeness score measuring coverage percentage. Calculate boundary adherence by dividing colored pixels inside boundaries by total colored pixels across the entire image and multiplying by 100, producing a score indicating what percentage of coloring stayed within designated lines.

4.Neatness and Consistency Feature Calculation: Extract neatness as a weighted combination of boundary adherence, fill density, and edge proximity. Calculate consistency by dividing the image into four quadrants, computing completeness for each quadrant, calculating the coefficient of variation across valid quadrants, and converting to a 0-100 score where lower variation indicates uniform effort throughout the template.

5.Template Complexity extraction: Count the number of detected shape regions from the contour list, calculate perimeter-to-area ratios for each contour and adjust based on the average perimeter-to-area ratio to produce a final 0-100 complexity score capturing task difficulty.

Data source 6:

1.Image Preprocessing and Standardization: Load and resize the tracing image to fix size, convert to both grayscale and HSV color spaces.

2.Line Accuracy Detection: Apply HSV thresholding with saturation range and value range to create a binary mask isolating the child's colored traced line, while simultaneously detecting the template path through dark line identification (HSV value) and Canny edge detection on the grayscale image to capture the expected route, then perform bitwise AND between the child's traced line mask and this corridor mask to count traced pixels inside the acceptable zone.

3.Line Completeness Detection: Extract the main path contour from the template detection mask to identify the expected tracing path, measure the total path length to determine the full distance the child should trace.

4.Complexity Detection: Analyze the template path contour to quantify task difficulty by calculating total path length, counting direction changes by examining angles between consecutive contour segments.

5.Consistency Detection: Divide the traced path into 4-6 logical segments by splitting the path contour at equal arc-length intervals or at identified waypoints where direction changes occur, creating distinct sections that can be evaluated independently. For each segment, calculate a local quality score by measuring that segment's completeness, corridor adherence, and smoothness.

Objective 4: IT22358066

Data source 7:

1. Key Features Evaluated:

To comprehensively capture handwriting characteristics, three complementary feature representations were extracted from each handwritten sample:

- **Structural Shape Features**: Extracted raw pixel intensities from grayscale images to capture the overall geometric structure, proportions, and spatial layout of handwritten characters. This representation preserves global shape information essential for distinguishing between different letter forms.
- **Skeletonized Stroke Patterns**: Derived thin, single-pixel representations of handwriting to evaluate the core trajectory and connectivity of strokes, independent of pencil thickness or writing pressure variations. This feature isolates the fundamental path of pencil movement, revealing consistent stroke patterns across different writing styles.
- **Gradient-Based Edge Features**: Captured using Canny edge detection to identify high-contrast boundaries and fine-grained structural details in character formation. This representation highlights critical transitional regions and sharp changes in writing direction, providing complementary information about stroke precision and letter boundaries.

2. Feature Transformation:

- **Multi-Channel Stacking**: The clean binary image, skeletonized stroke representation, and edge map were stacked into a single three-channel input array, enabling simultaneous learning of relationships between complementary feature modalities without discarding discriminative handwriting information.
- **Region-of-Interest (ROI) Extraction:** Bounding box coordinates were used to perform "Crop-to-Letter" transformations, ensuring that the feature set focuses strictly on the handwritten character region rather than irrelevant background noise.
- **Contrast Enhancement**: Contrast Limited Adaptive Histogram Equalization (CLAHE) was applied to the dataset to normalize illumination features across varying images and

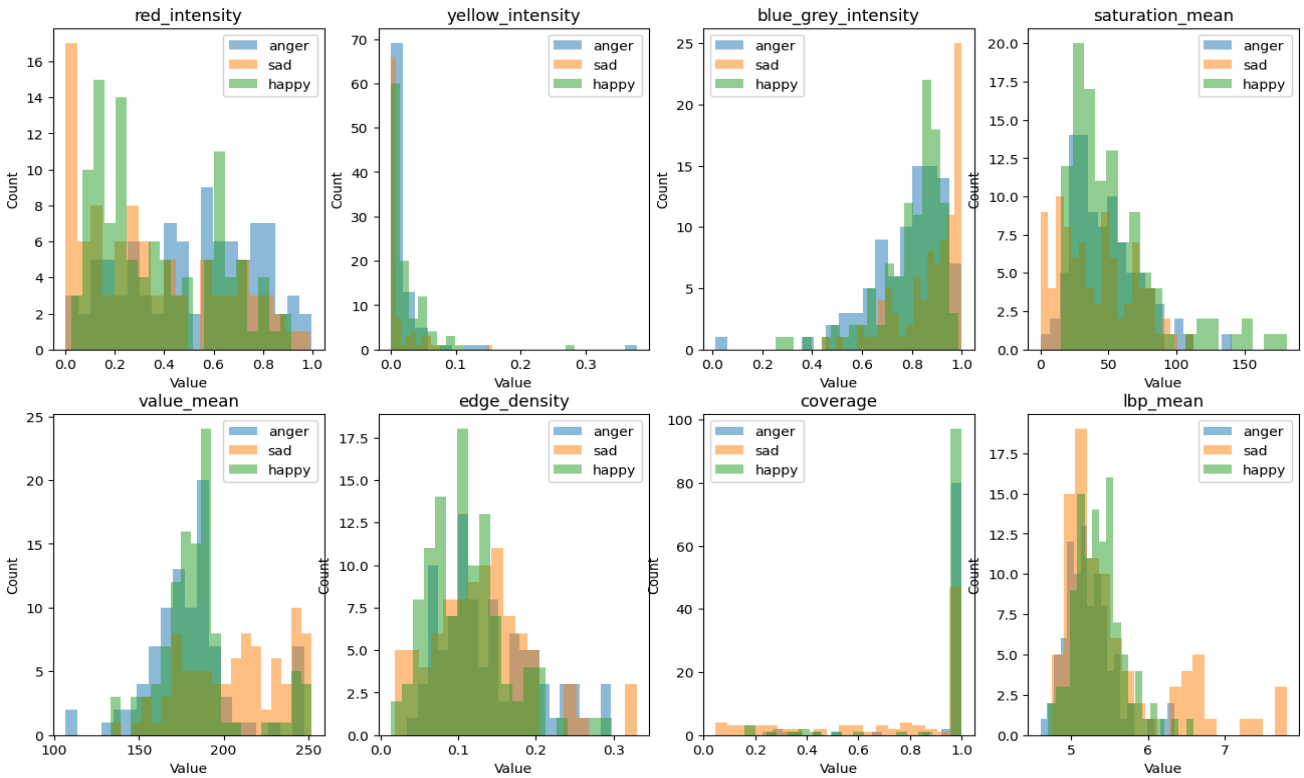
capture qualities. This reduces the impact of lighting inconsistencies on feature representation.

4. Modelling and Results

4.1. Key Insights:

Objective 1: IT22315250

Data source 1:
As per the psychological research papers that were done on preschool artwork and emotions, the same features were detected in the dataset as well. Below image shows how the dataset features align with the research paper findings [16-18],



The histograms represent how each feature is associated with the artwork labels of emotional categories anger, happy, sad.

Histogram 1-3: These plots show how different colors are more prominent in identifying certain emotions. According to researchs[16-18] anger is more associated with red, yellow is more associated with happy and

blue is more associated with sad. Even in the dataset features, we can see the slight difference of these categories.

Histogram 4: This shows the saturation mean of the art work that were collected. Research papers[16-18] show how high saturations often appears in happy labeled artwork and low saturation often appear in sad artwork. The histogram also reflects these research findings

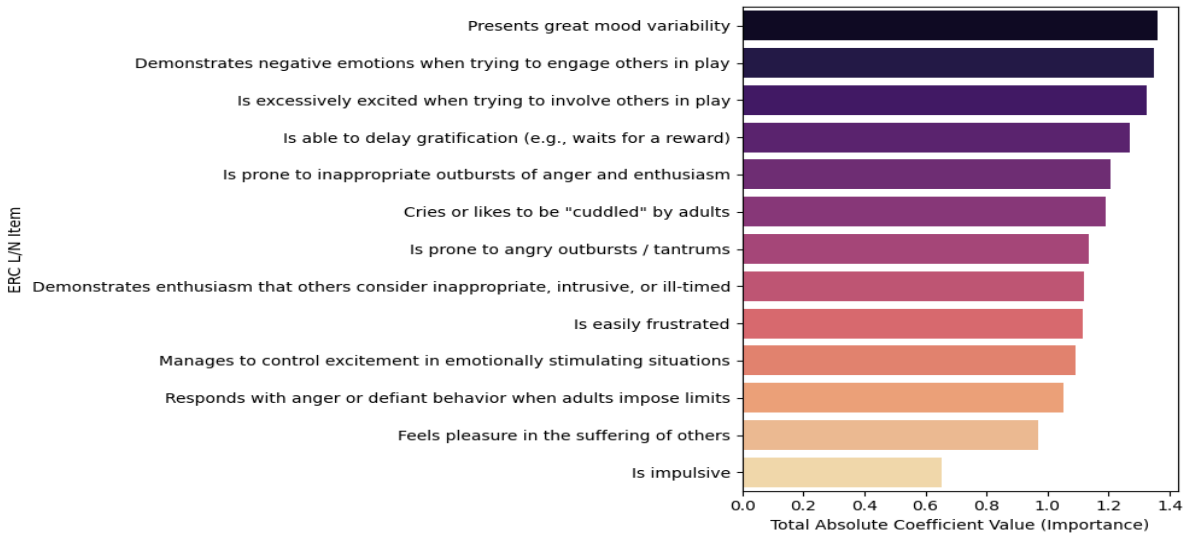
Histogram 5: The value_mean feature represents the average brightness of the artwork. Both happy and sad categories show high value_mean values. Happy drawings are bright due to the use of vivid colors, while sad drawings exhibit high brightness because of larger white background areas caused by limited drawing coverage. As a result, white regions increase the overall brightness despite the negative emotional expression.

Histogram 6: Edge density is more intense in in anger category because these art work tend to be more chaotic, forceful strokes which shows the inner turmoil a child is facing where as happy category will have smooth, curves reflecting a child's calm, happy states.

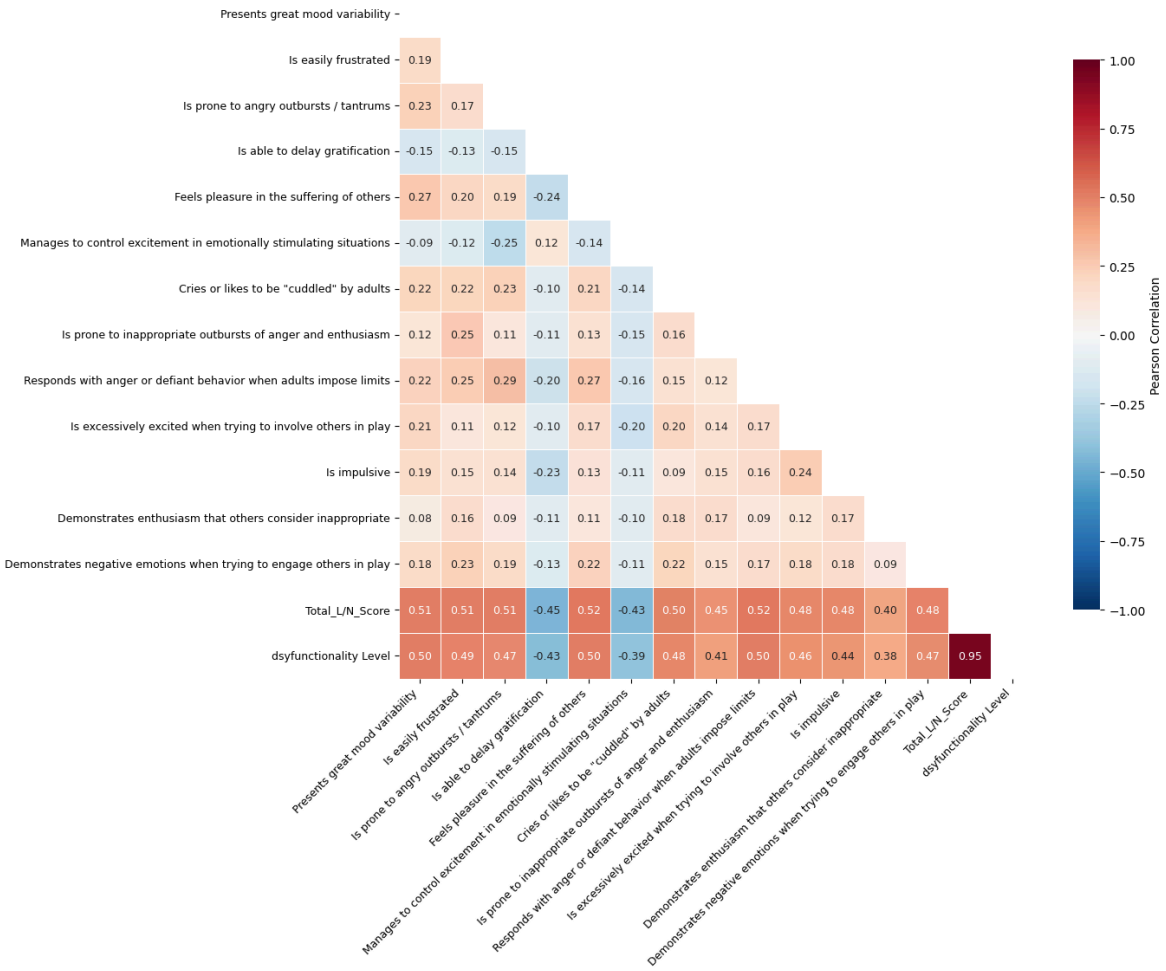
Histogram 7: This shows how much of the image is filled vs the negative space. Large coverage reflects more energy like happy and anger categories.

Histogram 8: Local Binary Pattern (lbp) mean captures the overall texture complexity. The higher the lbp mean values indicate more complex, irregular and detailed textures. This reflects that happy drawing generally tend to have smooth lines.
So these features show that we can identify emotions through the use of artworks

Data source 2:
The below image shows what features were given more importance to, these also match very closely with the research papers that were done for the emotional regulation checklist [15].



The emotional regulation checklist items had an expected correlation as research papers had mentioned. As the research papers had reflected both the 'is able to delay gratification' and 'manages to control excitement in emotionally stimulating situations' are having an inverse effect on the total scoring. The below image shows correlation matrix of the emotional regulation checklist and how it aligns with the research paper [15].



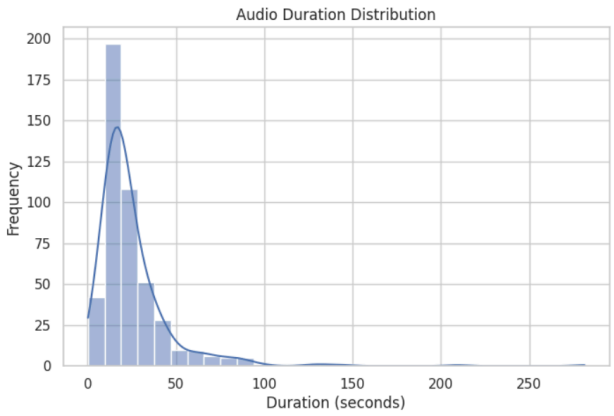
Objective 2: IT22345400

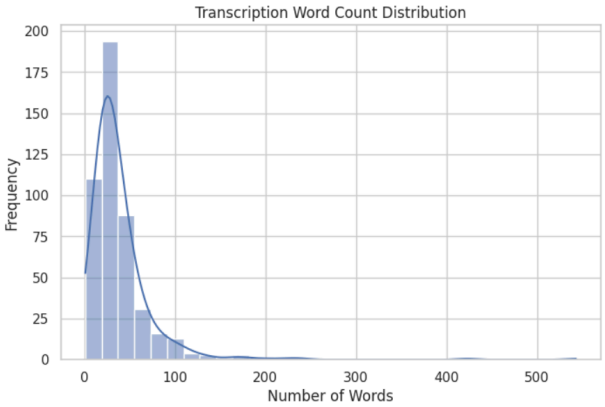
Data source 3:

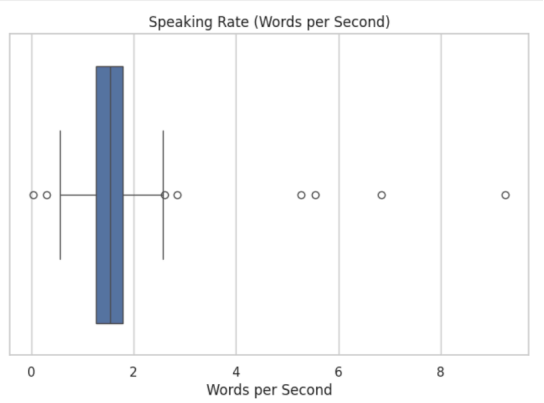
The primary objective of this study was to develop a localized automatic speech recognition (ASR) system for Sri Lankan children aged 4–6, addressing the unique challenges of child speech, including accent variation, pronunciation differences, and higher pitch compared to adult speech [19]. Existing ASR models, including OpenAI's Whisper-small English model, are primarily trained on adult voices and are often inadequate for accurately transcribing child speech, particularly from non-Western regions.

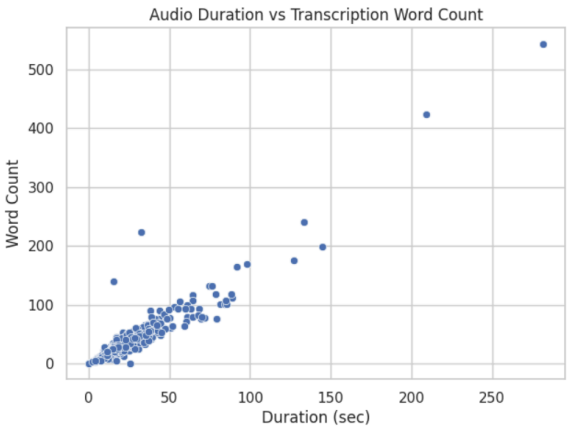
```
Dataset Overview
======================================
Total samples        : 467
Avg audio duration   : 25.24 sec
Avg words per sample: 39.2
Avg speaking rate    : 1.58 words/sec
```

Above shows the dataset overview of audio recordings.









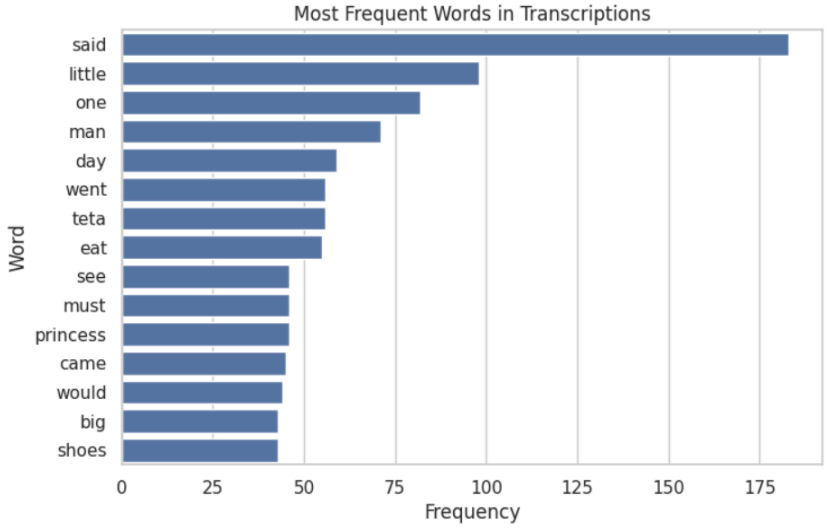
Audio Duration Distribution graph shows a Right-Skewed Distribution**.** The data follows a "long-tail" or

positively skewed pattern. The vast majority of your audio clips are concentrated in the shorter range (likely between 1 to 5 seconds), while only a few clips extend toward the maximum duration. The clips on the far right (the tail) represent outliers.

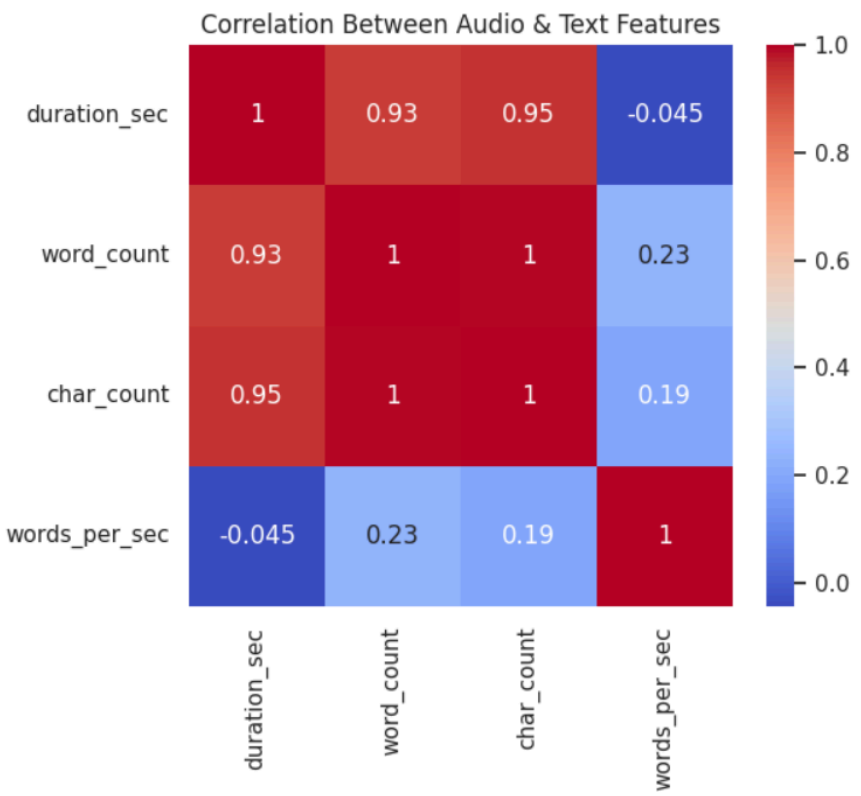
Transcription Word Count Distribution graph shows how much text corresponds to each audio file. Most transcriptions are short, peaking between 0 and 50 words. The shape of this graph closely follows the Audio Duration graph, which is a strong indicator of high-quality data labeling. There are rare instances of transcriptions exceeding 500 words, matching the audio outliers.

Speaking Rate (Words per Second Box Plot) illustrates the density and variance of how fast individuals are speaking in your dataset.The median speaking rate is approximately 1.5 to 1.8 words per second. The "box" (Interquartile Range) is quite narrow, suggesting a very consistent speaking pace across the bulk of the dataset. There are significant outliers on the high end, with some speakers reaching over 9 words per second. These likely represent segments with overlapping speech, background chatter, or technical errors in the transcription.

Scatter Plot shows a near-linear relationship between duration_sec and word_count (correlation coefficient of 0.93). The scatter plot confirms that as the audio plays longer, the word count increases predictably.



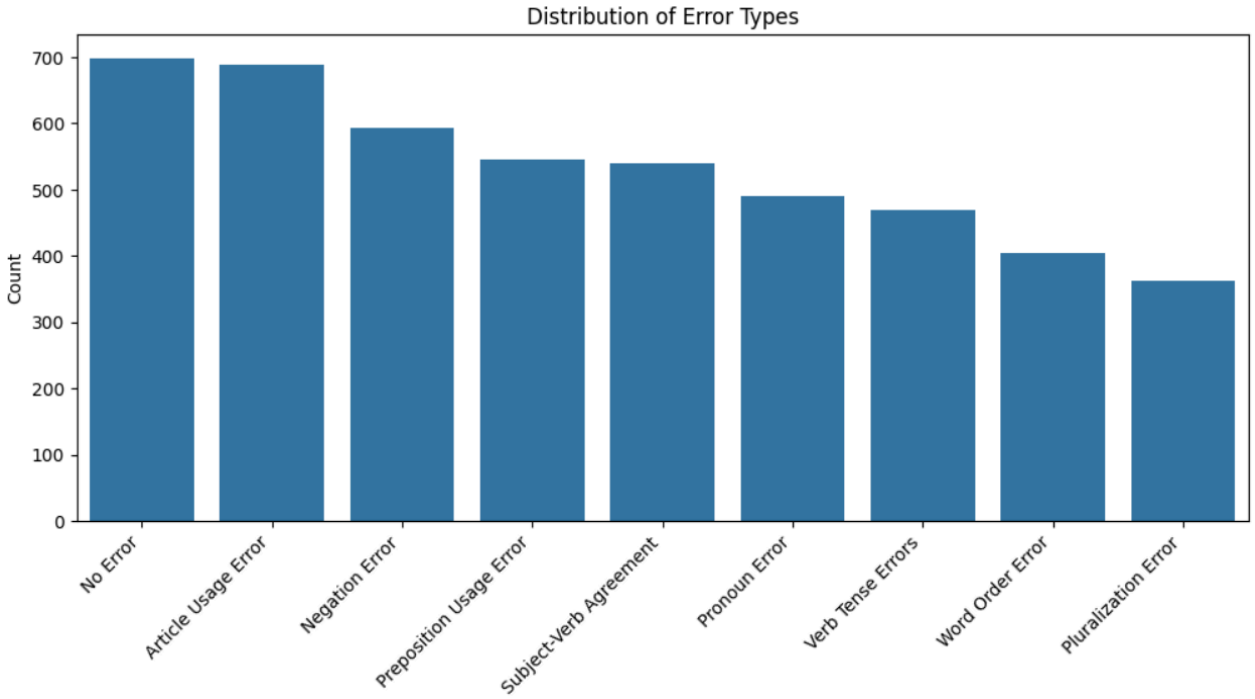
Above bar chart shows the most frequent words in transcriptions. “Said” is the most frequent word used.



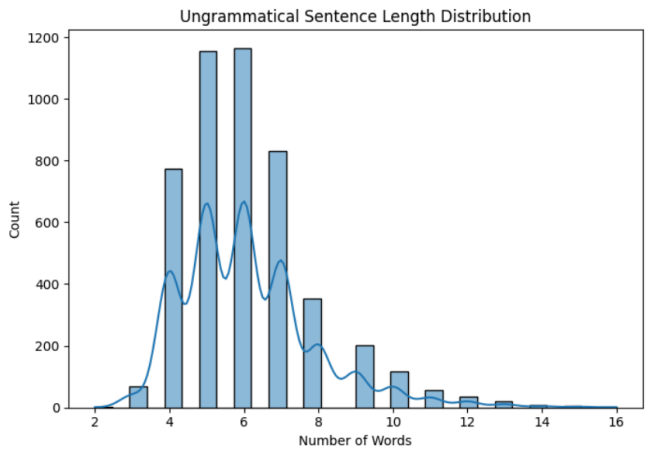
The above visualization is a heatmap showing the Pearson correlation coefficients between various dataset attributes. There is an extremely strong positive correlation between duration_sec and both word_count (0.93) and char_count (0.95). This indicates that as audio length increases, the transcript size increases linearly, suggesting a consistent data quality across the set. Word and Character Count have a perfect correlation of 1.0, which is expected as they both measure transcript length. Interestingly, words_per_sec has a near-zero correlation (-0.045) with duration_sec. This means that speakers do not necessarily slow down or speed up just because a clip is longer or shorter. The weak correlation between words_per_sec and other metrics suggests that the speaking pace is relatively uniform across the dataset, regardless of the total length of the file.

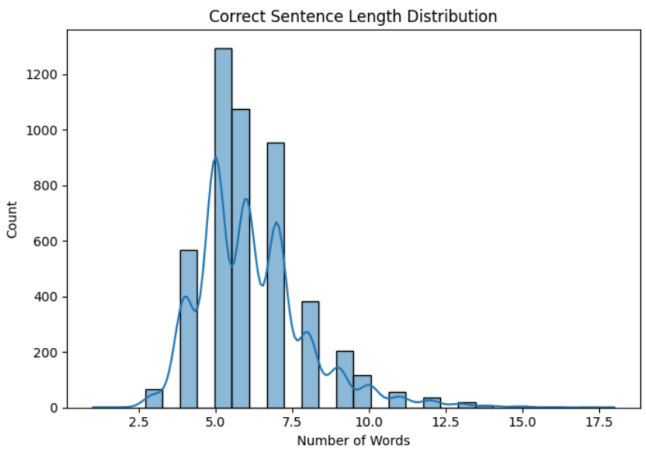
Data source 4:

The dataset is used to fine-tune grammar error detection and correction model that learns to transform ungrammatical child-like English sentences into standard English, while being conditioned on the error type. Below shows the analysis of the dataset:

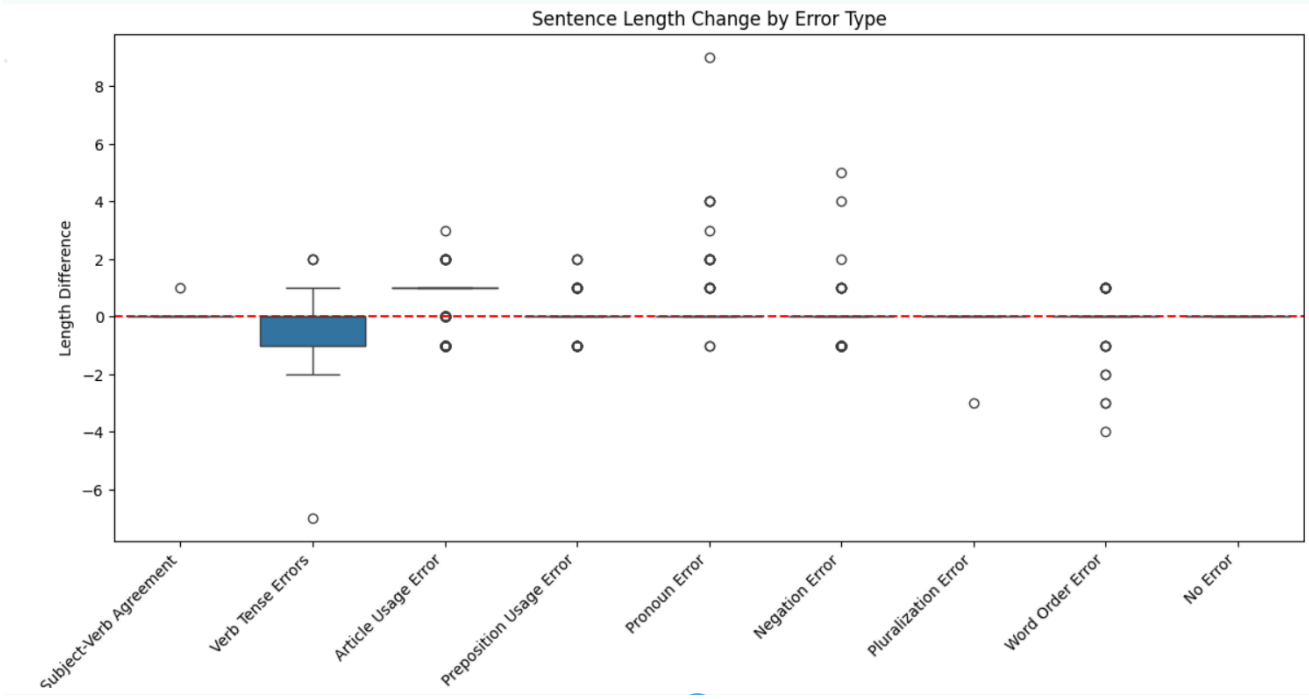


According to the above column chart, Highest error type is with no error category and lowest error type is pluralization errors.

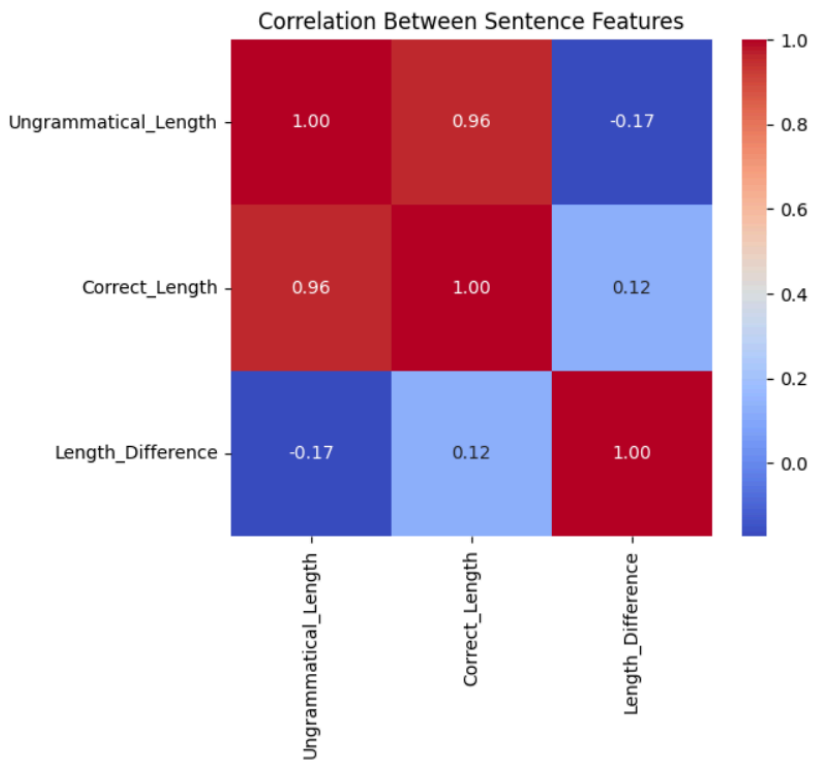




From the above histograms, both charts show a high density of sentences containing between 4 and 8 words, indicating this dataset is primarily focused on short, concise sentence structures rather than complex, multi-clause prose. The highest frequency (mode) is split between sentences of 5 and 6 words. The mode shifts slightly to a clearer peak at 5 words, which sees a count increase to over 1,300



Above box plot shows that the median length difference for almost every error category lies exactly on the 0.0 (red dashed line). This indicates that for the vast majority of cases in your dataset, correcting the error involves a one-to-one word replacement (e.g., swapping "run" for "runs") rather than adding or removing words. The No Error category is a flat line at zero with no outliers. This confirms the integrity of your "clean" data, showing that sentences labeled as correct are not being modified in length during the evaluation process.



There is an extremely high positive correlation of 0.96 between Ungrammatical_Length and Correct_Length. This indicates that the majority of corrections are minor edits (like word swaps) rather than significant sentence restructuring or paraphrasing.

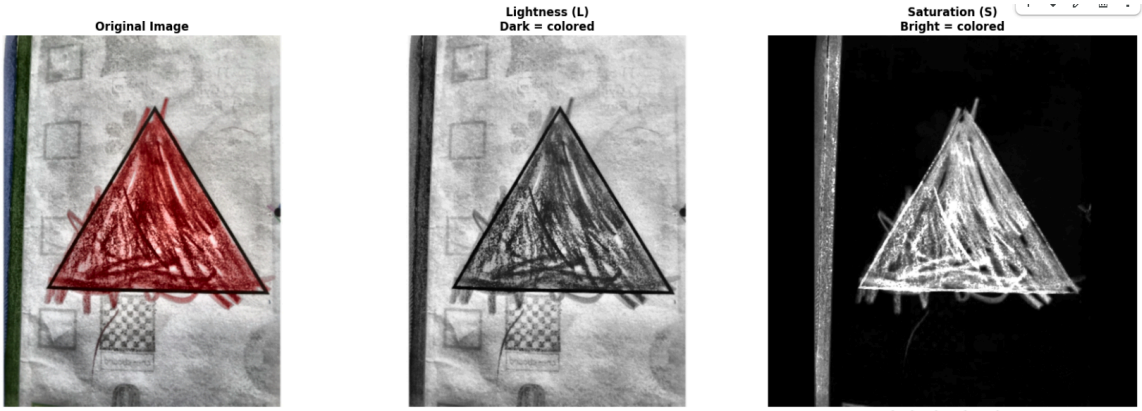
The correlation between Ungrammatical_Length and Length_Difference is very low at -0.17. This suggests that the number of words added or removed during correction is not strongly dependent on how long the original sentence was.

There is a weak positive correlation of 0.12 between Correct_Length and Length_Difference. This implies that as the final corrected sentence gets longer, it is slightly more likely that the correction involved adding words rather than removing them.

The negative -0.17 correlation between Ungrammatical_Length and Length_Difference indicates a slight tendency where longer original sentences may result in a smaller or negative length difference (pruning) compared to shorter ones.

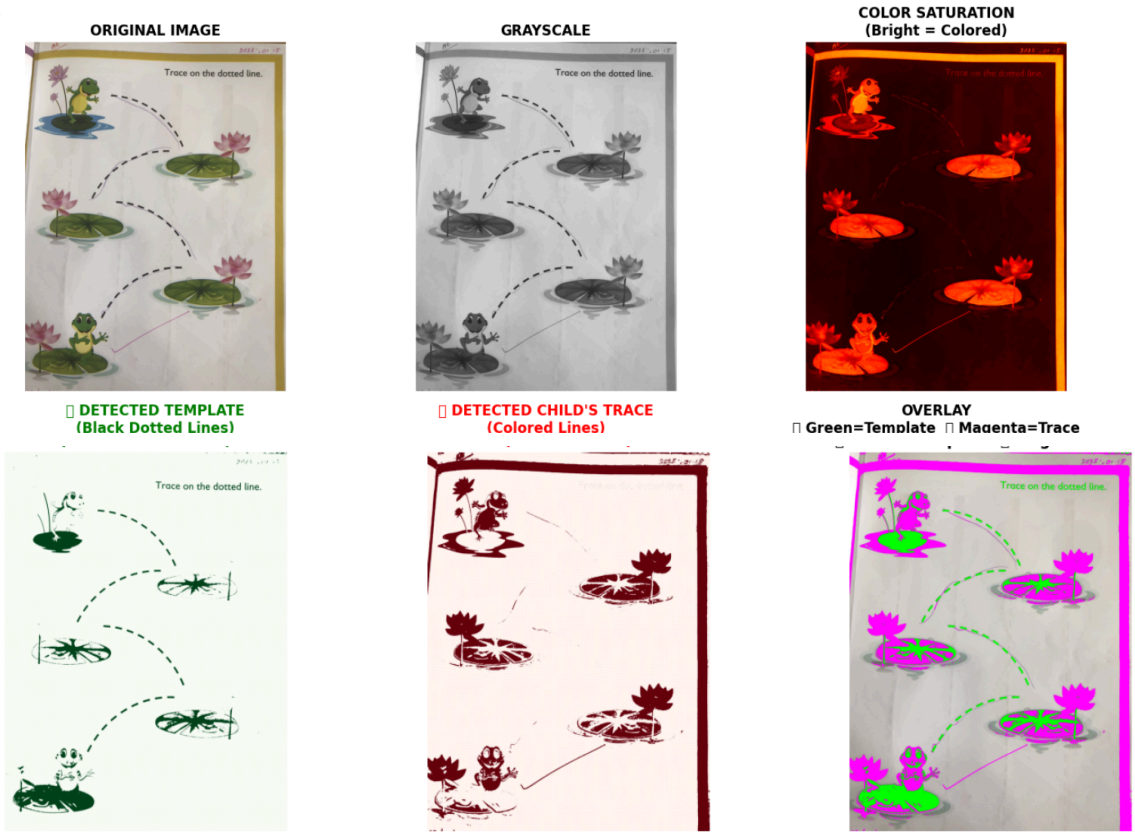
Objective 3: IT22640048

Data source 5: The system processes images to assess how well children have filled in predefined coloring templates, analyzing multiple quality dimensions including completeness, boundary adherence, neatness, and consistency.



From any given image, the system extracts quality feature values by first converting the image to HSV color space where the saturation channel isolates colored regions, while simultaneously detecting shape boundaries through combined edge detection and dark line identification to establish the expected fill area as the largest contour in the image. Using these masks, the system calculates five key numerical values: the completeness score measuring the percentage of expected area that contains color by dividing filled pixels by total expected pixels, the boundary adherence score measuring staying within lines by dividing colored pixels inside boundaries by total colored pixels, the neatness score combining boundary adherence plus fill density , the consistency score dividing the image into quadrants and calculating the coefficient of variation across regional fill percentages to detect uneven work, and the template complexity score based on the number of detected regions and their perimeter-to-area ratios to quantify task difficulty.

Data source 6: The system processes images to assess how well children have traced or followed predefined paths, analyzing quality dimensions including line accuracy, completeness, complexity, consistency.



This way identifies the trace line by detecting the template's expected path shown as black dotted lines through edge detection and dark line identification in the grayscale channel, while simultaneously identifying the child's actual traced line by isolating colored pixels in the saturation channel that represent where the child drew with their crayon or marker. By comparing these two lines, the system extracts five feature values: the completeness score calculated by measuring what percentage of the expected dotted path length has corresponding traced pixels nearby, the boundary adherence score determined by creating a corridor around the expected path and calculating the ratio of child's traced pixels that fall inside versus outside this corridor, the neatness score combining how closely the traced line follows the path centerline using distance transforms plus how smooth and continuous the traced line is without breaks or erratic movements, the consistency score dividing the path into segments between waypoints and measuring variation in tracing quality across segments to detect if the child maintained steady effort throughout, and the template complexity score based on the path's total length, number of curves, and sharpness of turns to quantify task difficulty. The overlay visualization displays the template path in green and the child's traced line in magenta, making deviations, gaps, and incomplete sections immediately visible for assessment.

Objective 4: IT22358066

Data source 7:

Feature Representation Analysis:

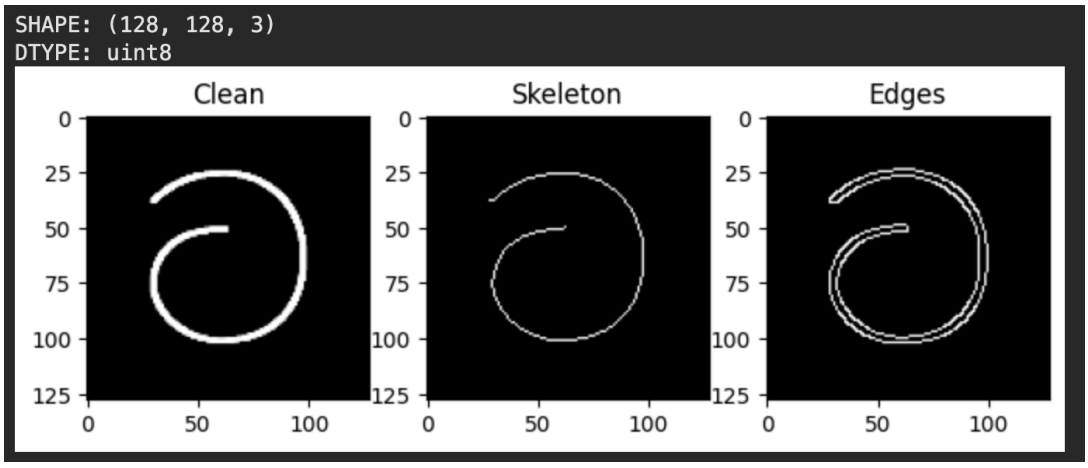


Figure 1

Figure 1 illustrates the extracted multi-representation handwriting features, including the clean binary image, skeletonized stroke representation, and edge-based representation.

The clean binary representation preserves the overall geometric structure of the handwritten character, enabling assessment of letter shape consistency. The skeletonized representation captures the core stroke trajectory and connectivity, independent of stroke thickness, allowing analysis of how accurately the child recalls the intended stroke order and form. The edge representation highlights fine-grained stroke boundaries and curvature, supporting discrimination between well-formed and distorted letter structures.

The consistency observed across these representations demonstrates that the extracted features effectively capture both global and local handwriting characteristics associated with memory recall performance.

Impact of Preprocessing on Handwriting Features:

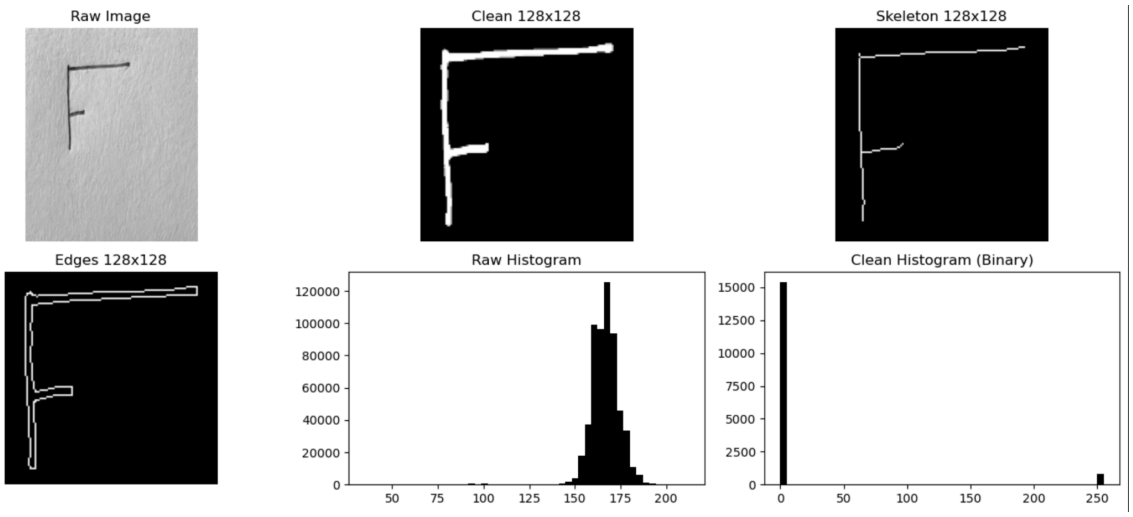


Figure 2

Figure 2 shows the transformation of a raw handwriting image into clean, skeleton, and edge representations, along with the corresponding intensity histograms before and after preprocessing.

The raw image histogram exhibits a wide intensity distribution caused by background texture, uneven lighting, and paper noise. After preprocessing and binarization, the clean histogram displays a highly concentrated intensity distribution, indicating successful isolation of handwriting strokes from background noise. This transformation confirms that preprocessing steps effectively standardize handwriting data while preserving meaningful stroke information required for reliable feature extraction.

The visual comparison between raw and processed images further demonstrates that noise removal and normalization significantly enhance the clarity of stroke patterns, which is critical for consistent handwriting analysis.

Interpretation in Relation to Memory Recall:

The extracted and transformed handwriting features reveal clear differences in stroke structure, connectivity, and spatial consistency, which are directly linked to memory recall ability. Children who accurately recalled learned letters produced handwriting with continuous, well-connected strokes and stable structural patterns, while incomplete recall resulted in fragmented or distorted stroke representations.

These results confirm that handwriting-based feature representations provide a reliable indirect measure of memory recall in preschool children.

4.2. Challenges Faced During Data Analysis:

Objective 1: IT22315250

Data labeling issues:

- After gathering art work of the preschool children, the art work was given to the art therapist for labeling of the data. Some drawings were easier to label because of domain knowledge but there were other data where the labeling was not very clear, so we avoided these types of artwork in order to have a quality dataset.
- We also found that with some of the artwork the preschool teacher had helped the student draw the art work. These types of artwork were also removed from the finalized dataset.
- For gathering data for the emotional regulation checklist, the information was limited to the students that were present on the days that we went to collect the information.

Objective 2: IT22345400

1. Acoustic and Physiological Outliers

Child speech is not just "high-pitched adult speech"; it is physiologically different due to shorter vocal tracts and undeveloped articulatory motor control.

- Pitch and Formant Shifts: Children aged 4–6 have a fundamental frequency (F_0) typically ranging from 250 Hz to 450 Hz, nearly double that of adults. This creates "spectral shifts" where the energy peaks (formants) are pushed higher, often confusing models like Whisper-small that expect adult spectral patterns.

- Intra-speaker Variability: A single child may pronounce the same word differently three times in a row. These "acoustic outliers" make it difficult for the model to establish a stable phoneme-to-text mapping.

- Duration Mismatches: As seen in earlier distribution plots, children's speaking rates are highly variable. They may speak very slowly when concentrating or rapidly when excited, creating temporal outliers that deviate from the "normal" speech rhythms found in Western adult datasets.

2. The "Preschool Environment" Challenge

Collecting data in a preschool is vastly different from a controlled studio environment.

- Non-Speech Noise (Missing Data/Corruption): Preschools are naturally loud. Background noise—such as other children playing, furniture moving, or fans—can overlap with the target audio. In some cases, the noise floor might be so high that the speech signal is effectively "missing" or "masked," leading to poor transcription accuracy.

3. Linguistic and Localized Variability

The "localized" aspect for Sri Lanka introduces a layer of complexity that standard English models cannot handle out of the box.

- Sri Lankan English (SLE) Accent: Localized pronunciations of specific vowels and consonants (e.g., the distinct "v" and "w" sounds or rhoticity patterns) differ from the US/UK English that *Whisper* was primarily trained on.
- Pronunciation "Errors": At age 4, phonological development is still ongoing. Children commonly substitute sounds (e.g., saying "wabbit" for "rabbit"). These are "labels" that are technically "correct" for the child but "outliers" for a standard English dictionary.
- Missing Vocabulary: Local names, places, or culturally specific objects (e.g., "poya," "hopper") may be absent from the *Whisper-small* vocabulary, causing the model to hallucinate more common Western words instead.

4. Computational and Data Limits

- Dataset Size: While 467 recordings is a significant manual effort and 6300 rows of grammar dataset, in the world of Deep Learning, it is considered a low-resource scenario. This limits the ability to train a model from scratch, forcing a reliance on "fine-tuning," which can lead to overfitting.

Objective 3: IT22640048

1.Limited Dataset Size for Deep Learning Applications

- The coloring dataset comprises 179 images and the tracing dataset contains only 62 images, both samples required for training robust deep learning models, creating high risk of overfitting and poor generalization even with data augmentation.

2.Image Quality and Capture Inconsistencies

- The images were different sizes that led to incorrectly identifying the template and colored regions tracing lines.

3.Identifying the given template along with child tracing and coloring

- The given child image through identifying the real template was difficult and had to use different techniques to identify accurately.

Objective 4: IT22358066

1. Data Gathering and Labeling issues

- Manual Annotation and Labeling Bottleneck: Annotating object images for the categorization activity was a time-consuming manual process. All images were carefully labeled and annotated to ensure ground-truth accuracy and reliable categorization, which required significant manual effort.
- Collection of Real-World Samples: Collecting a large number of student handwriting samples was difficult due to inconsistent lighting and writing environments across different preschool settings.
- Dataset Scarcity: There is a notable scarcity of publicly available datasets representing Sinhala handwriting of preschool children. Most existing handwriting datasets focus on adult handwriting or are primarily available for English language, making it difficult to source age-appropriate and culturally relevant data.

2. Computational and Practical Constraints

- Processing High-Resolution Image Data: The large volume of handwriting samples increased computational requirements during preprocessing, particularly for operations such as skeletonization and edge detection.

3. Data Preprocessing Challenges

- Noise and Background Artifacts: Handwriting images often contained unwanted noise such as shadows, uneven illumination, and background textures. These artifacts required careful preprocessing to ensure that only meaningful handwriting information was retained.

5. References

[1] J. Ebert et al., "Teaching Emotion and Creativity Skills to Preschoolers Through a Child Workshop," *The International Journal of Creativity & Problem Solving*, vol. 25, no. 2, pp. 23–35, 2015.

[2] Zhanna Iksanova, “A new method of developing emotional intelligence in preschool children through interactive digital facial expression recognition technology”, *in Proc. 2022 4th Int. Conf. Neurotechnologies and Neurointerfaces (CNN)* Kaliningrad, Russian Sep 2022, pp. 1-6, doi: 10.1109/CNN56452.2022.9912512

[3] A. L. Ruba and S. D. Pollak, "The Development of Emotion Reasoning in Infancy and Early Childhood," *Annual Review of Developmental Psychology*, vol. 2, pp. 503–531, 2020, doi: 10.1146/annurev-devpsych-060320-102556.

[4] S. Richard et al., "The development of specific emotion comprehension components in 1285 preschool children," *Scientific Reports*, vol. 12, no. 1, p. 19782, 2022, doi: 10.1038/s41598-022-22525-4.

[5] T. Kim et al., "Application of Deep Learning in Art Therapy," *International Journal of Machine Learning and Computing*, vol. 11, no. 6, pp. 496–500, 2021.

[6] H. Gomi et al., "Cross-Linguistic Study on Affective Impression and Language for Visual Art Using Neural Speaker," in *2022 10th International Conference on Affective Computing and Intelligent Interaction (ACII)*, 2022, pp. 1–6, doi: 10.1109/ACII55700.2022.9953880.

[7] K. A. R. Bucag, I. J. P. Casidsid, J. T. S. Enicola, and A. L. Ilao, "ADOMATH: An Android mobile game application for preschoolers," in *Proc. 2024 22nd Int. Conf. ICT and Knowledge Engineering (ICT&KE)*, Bangkok, Thailand, Nov. 2024, pp. 1–6, doi: 10.1109/ICTKE62841.2024.10787159.

[8] E. Sudarmilah, A. Susanto, R. Ferdiana, and N. Ramdhani, "Developing a game for preschoolers: What character, emotion and reward will tend to hack preschoolers?," in *Proc. 2015 Int. Conf. Data and Software Engineering (ICoDSE)*, Yogyakarta, Indonesia, Nov. 2015, pp. 1–6, doi: 10.1109/ICODSE.2015.7436977.

[9] M. K. Mokhtar, F. Mohamed, M. S. Sunar, M. A. M. Arshad, and M. K. M. Sidik, "Development of mobile-based augmented reality colouring for preschool learning," in *Proc. 2018 IEEE Conf. e-Learning, e-Management and e-Services (IC3e)*, Langkawi, Malaysia, Nov. 2018, pp. 1–6, doi: 10.1109/IC3e.2018.8632639.

[10] B. Parihar, S. Chhoriya, and R. Sangle, "Go-Brainous: An AI based educational app for kids," in *Proc. 2023 11th Int. Conf. Emerging Trends in Engineering & Technology – Signal and Information Processing (ICETET-SIP)*, Nagpur, India, Apr. 2023, pp. 1–6, doi: 10.1109/ICETET-SIP58143.2023.10151633.

[11] J. Johnson, "Effect of emotions on learning, memory, and disorders associated with the changes in expression levels: A narrative review," *Brain Circulation*, vol. 10, no. 2, pp. 134–144, Jun. 2024, doi: 10.4103/bc.bc_86_23.

[12] R. Ramírez-Esparza, A. T. García-Sierra, and A. K. Kuhl, "Beyond the 30-Million-Word Gap: Children's Conversational Exposure Is Associated With Language-Related Brain Function," Psychological Science, vol. 29, no. 4, pp. 700–710, Nov. 2017, doi: 10.1177/0956797617742725.

[13] R. Ramos et al., "Emotion Analysis in Children's Drawings Using Machine Learning Techniques," Research in Computing Science, 2024.

[14] A. Alshahrani, M. M. Almatrafi, and J. I. Mustafa, "A Children's Psychological and Mental Health Detection Model by Drawing Analysis based on Computer Vision and Deep Learning," *Engineering, Technology & Applied Science Research*, vol. 14, no. 4, pp. 15533–15540, 2024.

[15] Ş. Danışman, E. D. İman, Z. A. Demircan, and D. Yaya, "Examining the psychometric properties of the Emotional Regulation Checklist in 4- and 5-year-old preschoolers," *Electronic Journal of Research in Educational Psychology*, vol. 14, no. 3, pp. 534–556, 2016, doi: 10.14204/ejrep.40.15124.

[16] C. Damiano, P. Gayen, M. Rezanejad, A. Banerjee, G. Banik, P. Patnaik, J. Wagemans, and D. B. Walther, *"Anger is red, sadness is blue: Emotion depictions in abstract visual art by artists and non-artists," J. Vis.*, vol. 23, no. 4, Art. no. 1, 2023, doi: 10.1167/jov.23.4.1.

[17] D. Jonauskaite and C. Mohr, "Do we feel colours? A systematic review of 128 years of psychological research linking colours and emotions," *Psychonomic Bulletin & Review*, 2025, doi: 10.3758/s13423-024-02615-z.

[18] H.-C. Weng, L.-Y. Huang, L. Imcha, P.-C. Huang, C.-T. Yang, C.-Y. Lin, and P.-H. Li, *"Drawing as a window to emotion with insights from tech-transformed participant images,"* Sci. Rep., vol. 14, Art. no. 11571, May 2024, doi: 10.1038/s41598-024-60532-6.

[19] A. Ankita and S. Shahnawazuddin, "Developing children's ASR system under low-resource conditions using end-to-end architecture," Digital Signal Processing, vol. 146, p. 104385, Mar. 2024, doi: 10.1016/j.dsp.2024.104385.